

Unit - V

Q) Define Aerosols and its type. Write about formulation and its quality control tests?

A) Pharmaceutical Aerosol is defined as aerosol product containing API dissolved or suspended in a propellant or a mixture of solvent and propellant which on activation delivers the drug in desired form

Types of Aerosols

1) Metered Dose Inhalers (MDI)

- Drug is dissolved or suspended in non-aqueous propellants and packaged in specially designed pressurized container fitted with a metering valve
- A predetermined dose of Drug is ejected on actuation of the metering valve

2) Dry Powder Inhalers (DPI)

- Drug is inhaled as Fine Particles
- Activated whenever the Patient Inhales and therefore a Propellant is not required

3) Nebulizers

- Drug is suspended in aqueous phase and inhaled by facemask

→ Used for delivery of large therapeutic dose

4) Skin Sprays

→ Used for local effects eg Coolants, Analgesics

5) Dental Sprays

→ Used for local effects for teeth

* Components of Aerosols

1) Propellants (Heart of Aerosol)

→ Responsible for the maintenance of proper pressure

→ Expels the product when the valve is open

→ Used in metered dose inhalers formulation are gaseous at room temperature & pressure but immediately liquefied by decreasing temperature & increasing pressure

eg Tri-chloro-mono-fluoro methane (Oral & Inhalation)
Propane & Butane (Topical)

2) Container

Material used must be able to withstand pressure as high as 140 to 180 psig at 130°F

* Metals

• Tinplated Steel

• Aluminium

• Stainless Steel

* Glass (Pressure = 25 psig)

• Uncoated Glass

• Plastic Coated Glass

- 3) Actuator
→ Actuator helps in delivering the drug in desired form.

Types of Actuators

- 1) Spray Actuators 2) Foam Actuators
- 3) Solid Stream Actuators 4) Special Actuators

4) Valve

- Capable of delivering the content in the desired form (spray, foam, solid stream) & can deliver a given amount of medicament

Types of Valves

- 1) Continuous Spray Valve 2) Metering Valve

5) Therapeutic Agents

Solubility of therapeutic agents in the propellant system & size of particle of the dispersed particles are important physicochemical properties

Formulation of Pharmaceutical Aerosols

1) Product Concentrate

Product concentrate contains ingredients or mixture of active ingredients and other such as solvent, antioxidants and surfactants.

2) Propellant

Single or blend of various propellants is used to achieve desired solubility characteristics.

Quality Control Test of Aerosol

(1) Flammability and Combustibility

• Flame Expansion

- The product is sprayed for about 4 sec into a flame
- Depending on the nature of the formulation the flame extended
- The exact length is measured with scale

• Flash Point (Done by Standard Tag Open Cup Apparatus)

- The aerosol product is chilled to -25°F transferred to the test Apparatus
- Temperature Increased slowly & the temperature at which the vapour ignites is taken as Flash Point

2) Physiochemical Characteristics

1) Vapour Pressure

- The pressure can be measured simply with a pressure gauge or special equipment
- It is important that the pressure variation from containers be determined

2) Density

- The density of an aerosol system may be determined through a Hydrometer or a pycnometer

3) Moisture

- Karl Fisher Method or Gas chromatography has also been used

4) Aerosol Valve Discharge Rate (Performance)

- * This is determined by taking an aerosol product of known weight and discharging the contents (x) for a given time

↓
reweighed after given time (y)

- * Change in weight per time is its discharge rate expressed in gram/second

(2) Dosage with Metered Values

- Accurately weigh filled container and dispense (y) several dosage (x)
- The container is reweighed and difference in weight is divided by number of dose (z) dispensed

3) Net Contents

The uniformity in content of containers of a batch is determined by

↓
First weight the tared container & then container are filled and reweighed

↓
the difference in weight gives the Net Contents

4) Biological Tests

This test is performed to evaluate the therapeutic efficiency and toxicity of the aerosol products

(i) Therapeutic Activity

- * The dosage of product has to be determined for Inhalation aerosols & this must be related to particle size distribution
- * Topical preparation are applied to test areas in the usual manner and absorption of therapeutic agents can be determined

(ii) Toxicity

- Toxicity testing should include both topical and inhalation effects
- Aerosol applied topically may be irritating or chilling effect
- Inhalational toxicity must be done by exposing test animals to vapour & sprayed from an aerosol container.

Q) Discuss about materials used in packaging of pharmaceutical products along with their Quality Control Test?

A) Packaging is the science, art and technology of enclosing or protecting products for distribution, storage, sales and use

Types of Packaging:-

1) Primary Packaging

This is the first packaging envelope which is in touch with the dosage form eg Bottles, Ampoules

2) Secondary Packaging

This is consecutive covering or package which stores pharmaceutical packages in it for their grouping eg Boxes

3) Tertiary Packaging

This is to provide bulk Handling & Shipping of Pharmaceuticals from one place to another eg Containers, Barrels

1) Glass is the most commonly used in pharmaceutical packaging because it possesses superior protective qualities

→ It is chemically inert, impermeable, strong and rigid & has FDA clearance

Advantages

- They are transparent and can easily be labelled
- They are economical and have good protection power

Disadvantage

- They are ~~also~~ break easily
- They are heavier in comparison to plastic container

1) Type - I (Borosilicate Glass)

- Borosilicate glass is made up of combination of silica (SiO_2) (70 to 80%), Boron Oxide (7-13%), Sodium Oxide (Na_2O - 4 to 8%) & Aluminium oxide (2-8%)
- They are used for containing strong α , alkali & all types of solvents
- They are more chemically inert than soda lime glass

2) Type II (Treated Soda Lime Glass)

- These containers are made up of soda lime glass which treated with sulfur that neutralize alkali oxide and make glass more resistant
- They are used for storing aqueous solution having pH less than 7

Type III (Regular Soda Lime Glass)

- These are made up of soda lime glass which is untreated
- Type III not preferred for parenteral preparation

Type IV (General Purpose Soda Lime Glass)

- These glass are used for Tablet, Ointment & External liquids

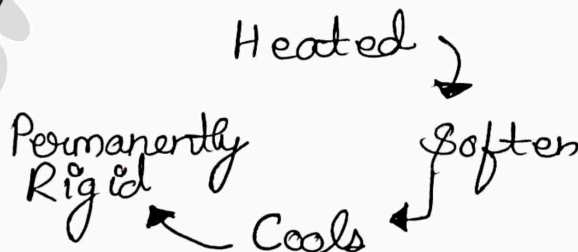
2) Plastic Containers

Plastic Container consist of Natural or Synthetic Polymer of High Molecular Weight that are shaped into shape by Heat and Pressure

(i) Thermosetting Type

- On Heating they are soften to viscous fluid which hardens again on cooling

Once Heated, thermosetting plastic can be moulded & shaped and pressed into shapes & once set they cannot be reheated since they are permanent by set



(ii) Thermoplastic type

On Heating they soften but solidifies (Harden) when cooled

eg Polyethylene, Nylon

3) Rubbers

- Rubbers are mainly used in the production of closure for vials, fluid bottles and transfusion
- Here are many types of rubber available which is used in closure production like Chloroprene Rubber, Silicon Rubber

4) Metals

- Metal are used in the construction of container
- Most commonly used metals for container are Aluminium, Stainless Steel & Tin

Quality Control for Glass

1) Powdered Glass Test

Principle :- This test is performed to estimate the amount of alkali leached from powdered glass

Procedure

Take containers & crushed in mortar & pestle

↓
Powdered glass is passed through sieve no 20 & 50

↓
Powder is washed with acetone and dried

↓
Take 10g of powder & place in a conical flask and add 50ml of purified water, cap the flask

↓
Place the container in autoclave 121°C for 30 min
cool the flask

↓
Separate the solution into another flask and add 5 drops of methyl red and titrate with $0.02\text{N H}_2\text{SO}_4$

Result

Glass pass the test → if the colour of test solution not change from pink to yellow

(ii) Hydrolytic Resistance Test

- Carry out the determination on unused containers to measure how much amount of alkali released Procedure when exposed to High Temperature & Humidity.

Fill Bottles or vials to 90% of their calculated overflow volume with distilled water & cover them with aluminium foil

↓
Place the containers in an autoclave, close the autoclave raise the temperature from 100°C to 121°C over 20 min

↓
Maintain a temperature 121°C for 60 min

↓
Reduce the Temperature 121°C to 100°C over 40 min

↓
Remove the container from autoclave and cool them

↓
Combine the liquids from containers in conical flasks and add methyl red

↓
titrate with 0.01N HCl

↓
Perform blank with water & the difference between the titration represents the volume

of HCl consumed by the test solution

3) Water Attack Test

Purpose :- To check whether the alkali leached or not from the surface of Soda Lime Glass

Procedure

Ring thoroughly with Purity Water

↓
Fill each container to 90% of its overflow capacity & is autoclaved at 121°C for 30 min

↓
Cool and liquid is decanted which is titrated with 0.02N Sulphuric acid using methyl red as an indicator

Result

The volume of sulphuric acid consumed is the measure of the amount of alkaline oxides present in the glass containers

4) Thermal Shock Test

This test is performed to evaluate that our glass containers can withstand sudden temperature change without physical damage or not

Procedure

Place sample container in upright position in tray & the tray immersed in Hot water for given time



Then the container transferred in cold water bath (45°C temp difference should be there)



The crack & examined on container

5) Internal Bursting Pressure Test

This ~~strong~~ test determine the internal pressure strength of container

Procedure

In this test American Glass Research Increment Pressure tester used

First of all Glass container filled with water and placed under Test chamber



The internal pressure automatically raised by time



And the Pressure is noted at last moment of Bursting of container

2) Quality Control Test for Plastic Container

1) Leakage Test

10 containers are randomly selected & filled with water & sealed



They are kept inverted at room temperature for 24 hours



Examine the leakage of container

2) Collapsibility Test

→ This test is performed on containers which are squeezed to remove contents

→ To check collapsibility, we use collapsibility tester equipment

→ This test ensure that the container yields at least 90% of its contents at the required flow rate and at normal temperature

3) Water Vapour Permeability Test

This test performed to measure the rate of transmission of water vapour through containers

Procedure

5 containers filled with water and Heat sealed with Aluminium foil / other



Then each container weighed and placed for 14 days at relative Humidity $60 \pm 5\%$ and at $20-25^\circ\text{C}$



After 14 days containers re-weighed & we check Quantity of lost H_2O

Q2) Write about the factors affecting selection of Packaging Material?

A) Pharmaceutical product packaging requires extreme care and the packaging must ensure maintenance of the efficacy of the product and its safe delivery.

→ Packaging material selection can be started with the knowledge of the products (regarding their stability & sensitivity)

Factors affecting selections of Packaging Material:-

1) Durability of Packaging Material

→ The packaging material should be of high quality and durability

→ If the formulation is having very short duration & shelf life then in such case less durable packaging material can be selected but it should be without compromising the quality of the material & vice versa

2) Cost of Material

→ There should be optimum balance between cost and quality of the material

→ Instead of changing the required material of packaging manufacturer can search for the cost effective source (supplier) of that material

3) Compatibility and Safety

- While choosing material for packaging manufacturer should confirm that the packaging material does not have any adverse effect on the formulation through chemical rxn, leaching of packaging materials, absorption or adsorption
- While performing compatibility study of formulation with packaging material sample of each drug which are selected for stability studies should be packaged in the same manner as they will be for marketing and distribution

4) Environment - friendly and legally compliant

- The material for construction of packaging should be environment friendly and legally sound with logistics and shipping laws & regulation
- The manufacturers also, make sure that the products you are buying should not infringe pre-existing, copyrights and patents

5) Degree of Protection Required

- The packaging material used for photosensitive material should possess the ability to protect the formulation from light
- This is also applicable to Hygroscopic, easily oxidized drug products & so on.

6) Convenience
How easy is it ^{for a} patient to use a formulation in terms of size, weight, method of opening or reclosing if appropriate during administration of the dosage form should be considered while choosing packaging material for a dosage form.

7) Printing
The legibility of printing is an important factor to consider while choosing and designing a package for formulation as it serves as a source of information and identification of a final product.

8) Type of Product

- The physical and chemical characteristics of the drug entity, the excipients, the formulation type of patient must be considered while dealing with the Pharmaceutical Product
- Apart from the properties of the drug, package style to attract patient and other legal requirements should also be considered during selection

9) Mechanical Properties

The materials used should possess sufficient mechanical strength to withstand while Handling, Filling closing and processing

→ Typical care is needed during transport, storage and also at the time of usage of product by the consumer especially in case of glass containers

1) Physical Properties of Packaging Material

- The material should be impermeable to any possible contaminants for example solids, liquid, gases vapours or microorganisms
- The container must be able to withstand Heat if the processing includes sterilization

Q) Preparation and Formulation of Cold Cream, Tooth-paste & Vanishing Cream?

A) Cold creams are water in oil or oil in water type emulsions added with certain fats such as Beeswax and perfuming agents

→ These are applied on skin to provide smoothness and named so due to cooling effect on application

Formulation of Cold Cream

Formula	Quantity for 100gm
White Beeswax (emollient)	20gm
Mineral Oil (lubricant)	50gm
Distilled Water (vehicle)	28.2gm
Borax (Buffer)	• 7gm
Perfume (Odour)	• 5gm

Preparation of Cold Cream

1) Prepare the Oil Phase

→ Melt Bee's wax and other oils like almond or mineral oil together in a container at a temperature of $75-80^{\circ}\text{C}$

2) Prepare the Aqueous Phase

→ Dissolve Borax in distilled water and Heat to the same temperature as the oil phase

3) Combine the Phase

Slowly add the aqueous phase to the oil phase while stirring constantly until the mixture is uniform

4) Cool

Continue stirring the mixture as it cools to room temperature to ensure proper emulsification

5) Add perfume

Add a suitable perfume to the preparation

Vanishing Cream are oil-in-water type of emulsion when applied on skin they spread easily and disappear

→ They are used to Hold powder on the skin and to improve adhesion

Formulation of Vanishing Cream

Formula	Quantity for 100gm
Stearic acid (Maintain cream consistency)	15%
Potassium Hydroxide (To neutralize free fatty acid)	7%
Glycerin (Enhance spreadability)	5%
Water (vehicle)	79%
Perfume, Preservative	q.s.

Preparation of Vanishing Cream

Melt Stearic acid in water bath

↓
Dissolve KOH in water and glycerin and Heat to 75°C

↓
Slowly add the aqueous phase to the Melted Stearic acid

↓
Add perfume when the mixture reaches 40°C

Tooth Paste are oral preparation meant for application on teeth for cleaning action and for certain therapeutic purpose

Formulation of Tooth Paste

1) Abrasive Agents

Abrasive or Polishing Agents are used to polish the teeth and to remove food particle adhered to surface of the teeth

eg Dicalcium Phosphate

2) Foaming Agents / Surfactants

They enhance cleaning property of tooth paste by reducing the interfacial tension b/w adhered material and enamel of teeth

eg Sodium Lauryl Sulphate

3) Humectants

- They are used to prevent rapid drying of paste
- They also provide plasticity to tooth paste
eg Glycerine, Propylene Glycol

4) Binding Agents

- They are used to hold the solid and liquid components together to form smooth paste
eg Cellulose Ether, Carboxymethyl Cellulose

5) Sweetening Agents

- They are used to improve the sweetening properties and cover the bitter taste of other ingredients

6) Flavouring Agents

- These are mixture of edible oils having spearmint & peppermint oil is the major constituents

7) Colouring Agents

- They are mainly used to influence consumer

8) Preservatives

- These are added to maintain stability of preparation and improve shelf life

Preparation of Tooth Paste

All the solid ingredients except surfacants are mixed together in a dry mixer



Liquid ~~the~~ components such as Humectant and water are gradually added to the dry mix



Mixing is carried out until a smooth paste is formed



Remaining Ingredients like surfacants and the flavouring agents are added to the Homogenous Paste